

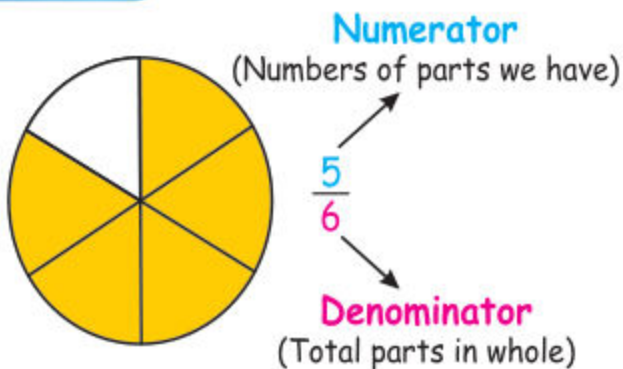
Fractions:



$\frac{1}{4}$ is a fraction or a fractional number.
The number at the top is called a Numerator.

The number at the bottom is called a Denominator.

Example:



Here fraction $\frac{5}{6}$ represents the shaded parts of the circle, indicating that 5 of the 6 parts are shaded. The numerator in this case is 5, while the denominator is 6.

Unit Fractions:

Fractions, where the numerator is '1', are called unit fractions.



In this example, the apple is divided into two equal parts.

Each part represents $\frac{1}{2}$.

$\frac{1}{2}$ is a unit fraction.



Here $\frac{1}{3}$ represents one third part of the apple. $\frac{1}{3}$ is a unit fraction, as the numerator is 1.

More examples of unit fractions are given below.



$$\frac{1}{4}$$



$$\frac{1}{5}$$



$$\frac{1}{6}$$



Activity

$\frac{1}{10}$ is a unit fraction. How many $\frac{1}{10}$ are in the figure show below:



Non-Unit Fractions:

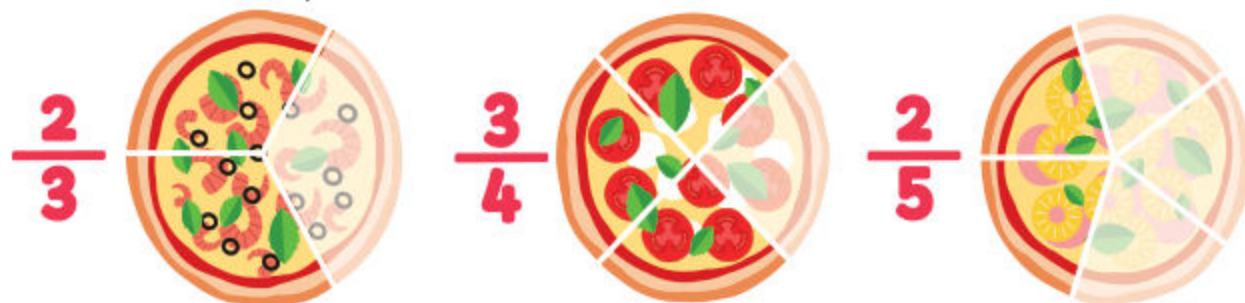
Fractions where the numerator is greater than "1" are called non-unit fractions.



$\frac{2}{3}$ are pink

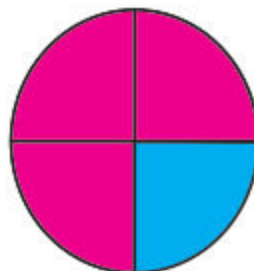
$\frac{2}{3}$ is not a unit fraction.

Other examples of non-unit fractions

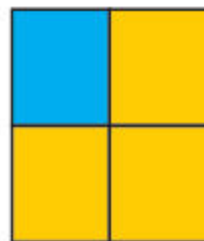


Exercise 1

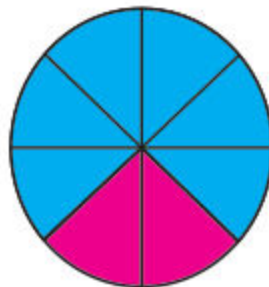
- i) Which colour of this circle represents a unit fraction?



- ii) Which colour of this square represents a non-unit fraction?



- iii) What fraction of this circle is blue?



- iv) Which fraction represents a unit fraction?

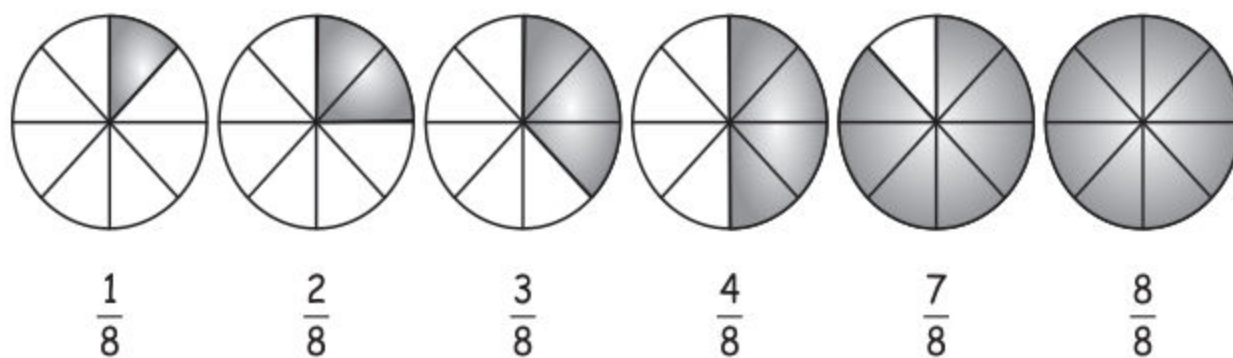


- v) Circle the non-unit fractions.

 $\frac{1}{4}, \frac{2}{3}, \frac{3}{5}, \frac{1}{8}, \frac{1}{6}, \frac{1}{7}$

Like Fraction:

When two or more fractions have the same denominator, they are called *like fractions*. The fractions given below are examples of like fractions since the denominator in these examples is 8.



Example:



The fractions $\frac{2}{5}$, $\frac{3}{5}$ and $\frac{1}{5}$ are the example of like fractions since the denominator, in these example is "5".



Activity

What is the denominator in the given fractions?



$$\frac{1}{5}$$



$$\frac{2}{5}$$



$$\frac{3}{5}$$



$$\frac{4}{5}$$

Other Examples of like fractions are:

(a) $\left(\frac{2}{9}, \frac{3}{9}, \frac{5}{9}, \frac{9}{9}\right)$

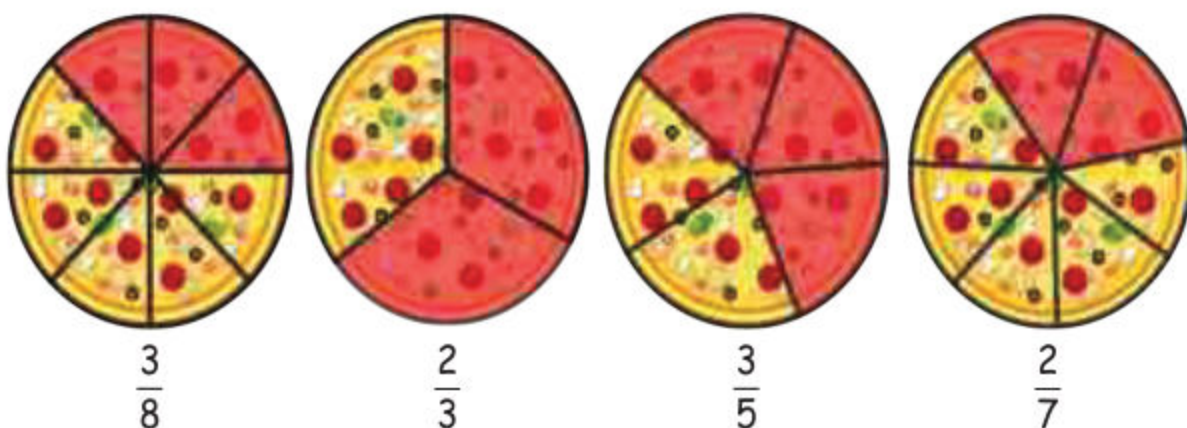
(b) $\left(\frac{1}{7}, \frac{2}{7}, \frac{4}{7}, \frac{5}{7}, \frac{7}{7}\right)$

Unlike Fractions:

Two or more fractions with different denominators are called unlike fractions.

Example:

In this example, $\frac{3}{8}, \frac{2}{3}, \frac{3}{5}$ and $\frac{2}{7}$ are unlike fractions because they do not have the same denominators.



Activity

Which figure is an example of an unlike fraction?

Figure A

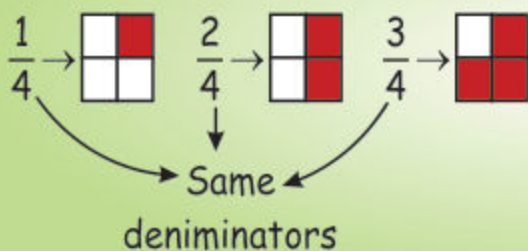
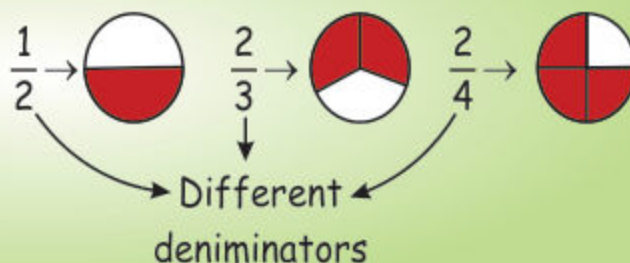


Figure B



More examples of unlike fractions are:

(a) $\left(\frac{1}{2}, \frac{1}{4}, \frac{2}{3}, \frac{5}{6}\right)$

(b) $\left(\frac{1}{9}, \frac{2}{7}, \frac{3}{4}, \frac{2}{5}\right)$

Exercise 2

1. Write whether the following sets of fractions are like or unlike.

i) $\frac{4}{7}, \frac{3}{7}$ ii) $\frac{5}{6}, \frac{13}{9}$

iii) $\frac{8}{5}, \frac{1}{2}$ iv) $\frac{4}{3}, \frac{5}{3}$

v) $\frac{1}{4}, \frac{5}{4}$ vi) $\frac{1}{12}, \frac{6}{11}$

vii) $\frac{9}{5}, \frac{7}{6}$ viii) $\frac{13}{13}, \frac{13}{13}$

2. Circle the pairs of like fractions.

i) $\frac{8}{11}, \frac{5}{11}$ ii) $\frac{7}{4}, \frac{2}{3}$

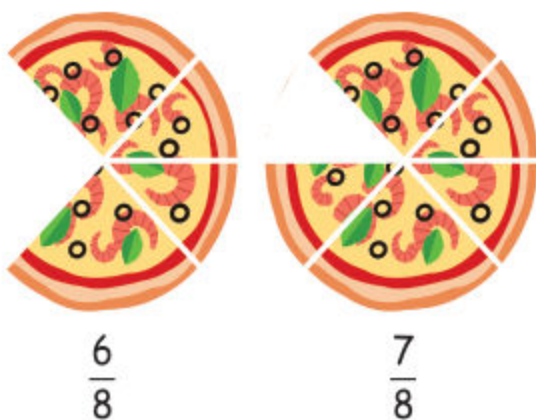
iii) $\frac{1}{6}, \frac{7}{6}$ iv) $\frac{9}{7}, \frac{9}{8}$

3. Circle the pairs of unlike fractions.

i) $\frac{6}{5}, \frac{8}{7}$ | ii) $\frac{1}{9}, \frac{3}{2}$ | iii) $\frac{10}{9}, \frac{2}{9}$ | iv) $\frac{3}{14}, \frac{7}{15}$

Comparing Fractions:

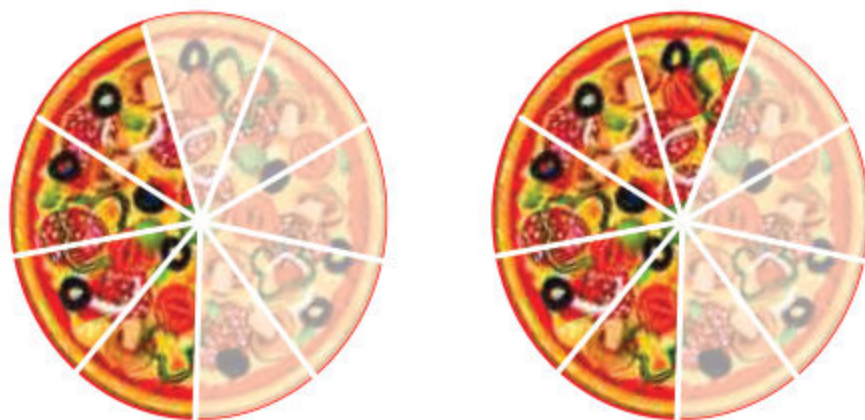
By comparing the numerators of any two like fractions, we can tell which is larger and which is smaller fraction. The greater the numerator, the larger the fraction.



If Ali ate six parts of a pizza, which was divided equally into eight parts, and his friend ate seven parts of the same size pizza. In this case, the denominator 8 remains the same, and the numerator 7 is greater than 6, so $\frac{7}{8}$ is greater than $\frac{6}{8}$.

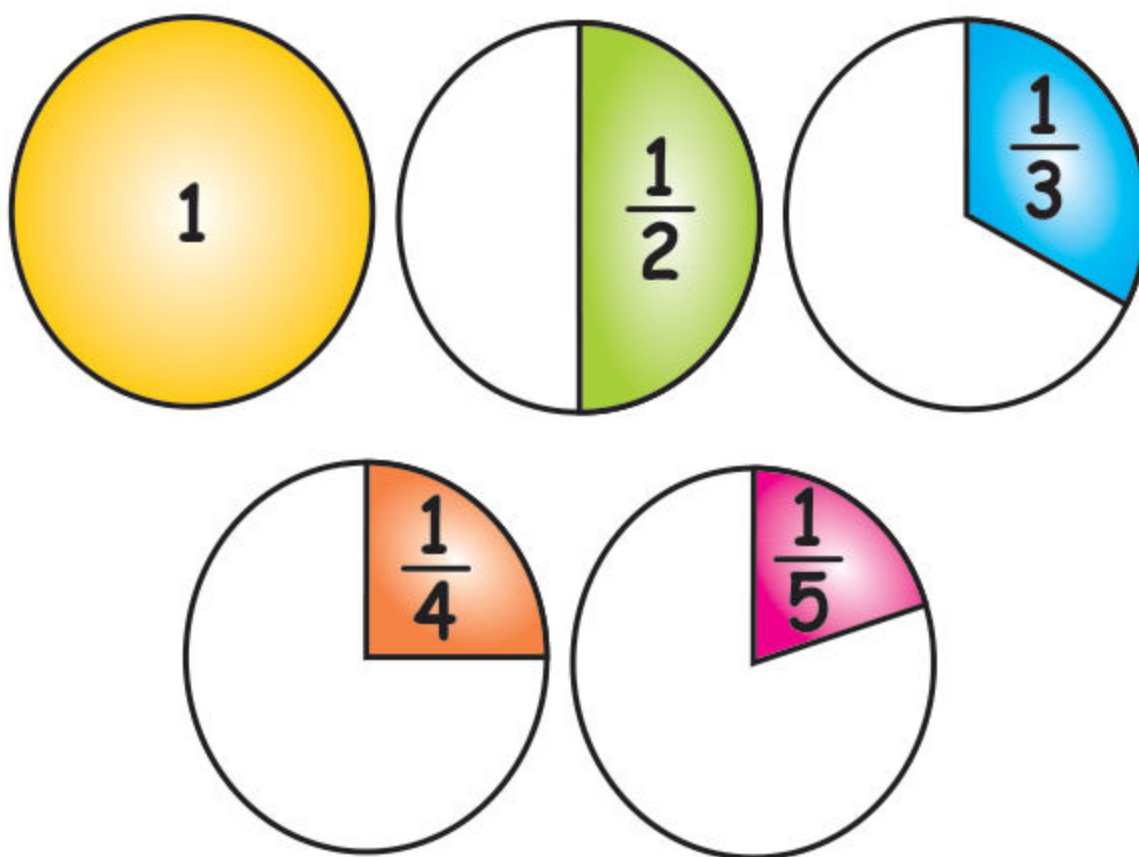
Example:

In the example below, $\frac{5}{9}$ is greater than $\frac{4}{9}$ (because 5, the numerator, is greater than 4, the numerator, while the denominator remains the same in both fractions).



Additionally, increasing denominator mean that we increase the number of parts needed to make a whole.

$\frac{1}{2}$ is greater $\frac{1}{3}$; and $\frac{1}{3}$ is greater than $\frac{1}{4}$ etc.



When the numerator is the same, the fraction with the greater denominator is smaller. We can say $\frac{1}{4}$ is smaller than $\frac{1}{2}$. $\frac{1}{5}$ is smaller than $\frac{1}{4}$.

Remember:

- With the same denominator, a fraction of greater numerator means a greater fraction.
- With the same numerator, a fraction of the smaller denominator means a greater fraction.
- When the fractions have the same denominator, you can write them in ascending order (from greater to smaller) or descending order (from smaller to greater).

Exercise 3

1. Choose any two fractions from the examples given below and compare which is a smaller fraction and which is a greater fraction.



$$= \frac{1}{2}$$



$$= \frac{1}{3}$$



$$= \frac{1}{4}$$



$$= \frac{1}{5}$$



$$= \frac{1}{6}$$



$$= \frac{1}{7}$$



$$= \frac{1}{8}$$

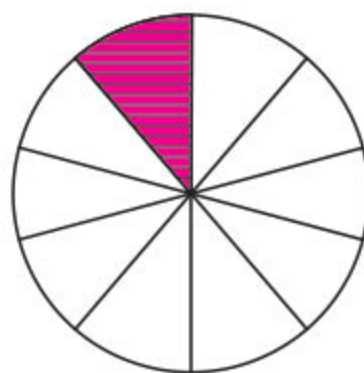
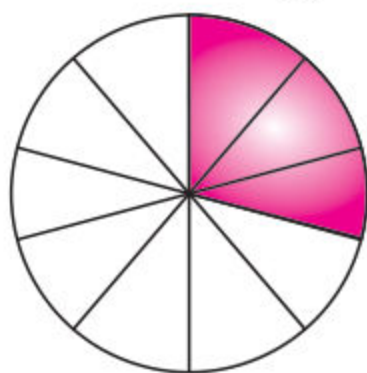


$$= \frac{1}{9}$$



$$= \frac{1}{10}$$

2. Which one is greater $\frac{1}{10}$ or $\frac{3}{10}$

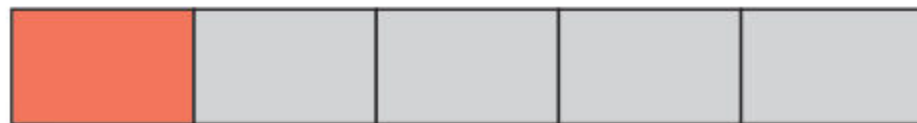


3. Tick the smaller fraction.

$$\frac{1}{4}$$



$$\frac{1}{5}$$

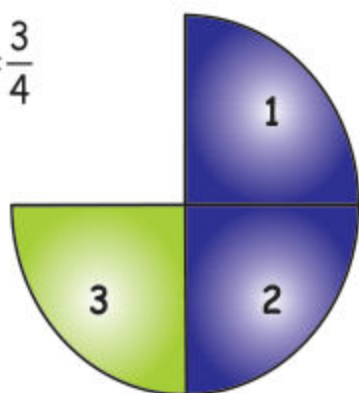


4. Reema ate $\frac{3}{5}$ of the pizza, and Anum ate $\frac{4}{5}$ of the pizza. Who ate more pizza?
5. Raj lives $\frac{1}{4}$ km from the school, and Pirya lives $\frac{2}{4}$ km from the school. Who lives closest to the school?
6. Danial's garden has $\frac{2}{10}$ of flowers and $\frac{5}{10}$ of vegetables. Are there more flowers or vegetables in his garden?

Addition of Like Fractions:

While adding the two or more like fractions, we add the numerators of the fractions and do not change the denominator.

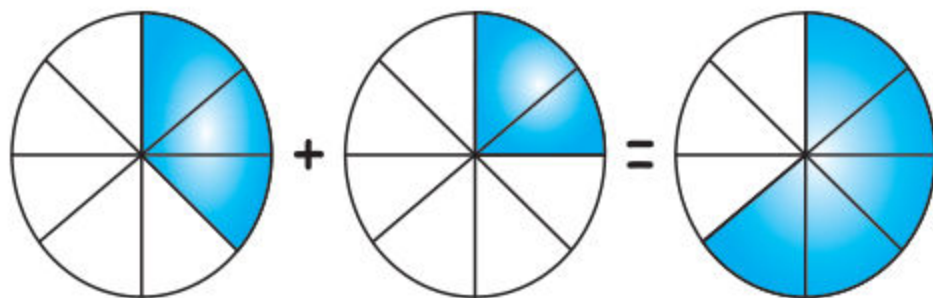
$$\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$$



For example, Anisa ate $\frac{1}{4}$ of a loaf of bread, and her friend ate $\frac{2}{4}$ of the loaf. To find the total, we added the numerators of the two fractions, so together, they ate $\frac{3}{4}$ of the bread.

Example:

$$\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$$



To add like fractions, we add the numerators and write the sum over the denominator.



Activity

Find $\frac{4}{9} + \frac{3}{9}$

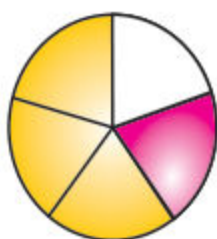
Since the denominators are the same, add the numerators.

$$\frac{4}{9} + \frac{3}{9} = \frac{4+3}{9} = \frac{7}{9}$$

Exercise 4

1. Add the like fractions as given below:

1)



$$\frac{3}{5} + \frac{1}{5} = \frac{\quad}{5}$$

$$\frac{1}{5} + \frac{3}{5} = \frac{\quad}{5}$$

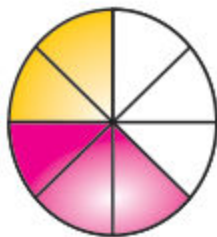
2)



$$\frac{5}{7} + \frac{1}{7} = \frac{\quad}{7}$$

$$\frac{1}{7} + \frac{5}{7} = \frac{\quad}{7}$$

3)



$$\frac{2}{8} + \frac{3}{8} = \frac{\quad}{8}$$

$$\frac{3}{8} + \frac{2}{8} = \frac{\quad}{8}$$

4)



$$\frac{3}{10} + \frac{4}{10} = \frac{\quad}{10}$$

$$\frac{4}{10} + \frac{3}{10} = \frac{\quad}{10}$$

Subtraction of Like Fractions:

To subtract two like fractions, we subtract the numerators and do not change the denominator.

Example 5:

Alisha got $\frac{3}{4}$ of a pizza and ate $\frac{2}{4}$ of it.

How much pizza is left?

To solve this problem, we subtract 2 from 3, which is 1.

$$\frac{3}{4} - \frac{2}{4} = \frac{1}{4}$$

Hence, the remaining pizza is $\frac{1}{4}$.



$$-\frac{1}{4}$$



Activity

1) - =
 $\frac{3}{6} - \frac{2}{6} =$

2) - =
 $\frac{5}{10} - \frac{2}{10} =$

3) - =
 $\frac{4}{5} - \frac{1}{5} =$

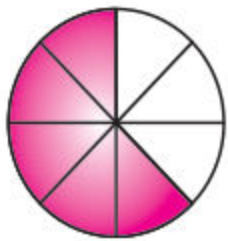
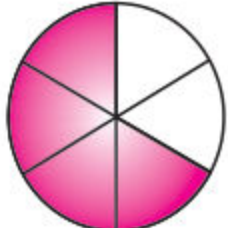
4) - =
 $\frac{6}{8} - \frac{4}{8} =$

5) - =
 $\frac{8}{12} - \frac{5}{12} =$

6) - =
 $\frac{2}{4} - \frac{1}{4} =$

Exercise 5

Solve the questions given in the table.

Fraction	Problem	Subtraction of Fractions
1) 	Ali had $\frac{5}{8}$ of pizza. He ate $\frac{2}{5}$ of the pizza. What fraction is left?	$\frac{5}{8} - \frac{2}{8} = \frac{3}{8}$
2) 	Amina had $\frac{4}{6}$ of the pizza. She ate $\frac{1}{6}$ of the pizza. What fraction is left?	$\frac{4}{6} - \frac{1}{6} = \frac{3}{6}$
3) 	Sapna had $\frac{4}{5}$ of the pizza. She ate $\frac{3}{5}$ of it. What fraction is left?	$\frac{4}{5} - \frac{3}{5} = \frac{1}{5}$

4. Solve the following

i) $\frac{5}{8} - \frac{3}{8}$

ii) $\frac{2}{3} - \frac{1}{3}$

iii) $\frac{6}{6} - \frac{3}{6}$

iv) $\frac{1}{2} - \frac{1}{2}$

v) $\frac{2}{10} - \frac{1}{10}$

vi) $\frac{5}{7} - \frac{4}{7}$

Unit 5

ALGEBRA (PATTERNS)

Student Learning Outcomes:

At the end of the chapters, Students will be able to:

- ✓ Describe and extend a repeating non numerical pattern with two core elements through pictorial and/or visual representations.



- ✓ Extend and create patterns in a variety of ways e.g. using 2D and 3 D shapes.

